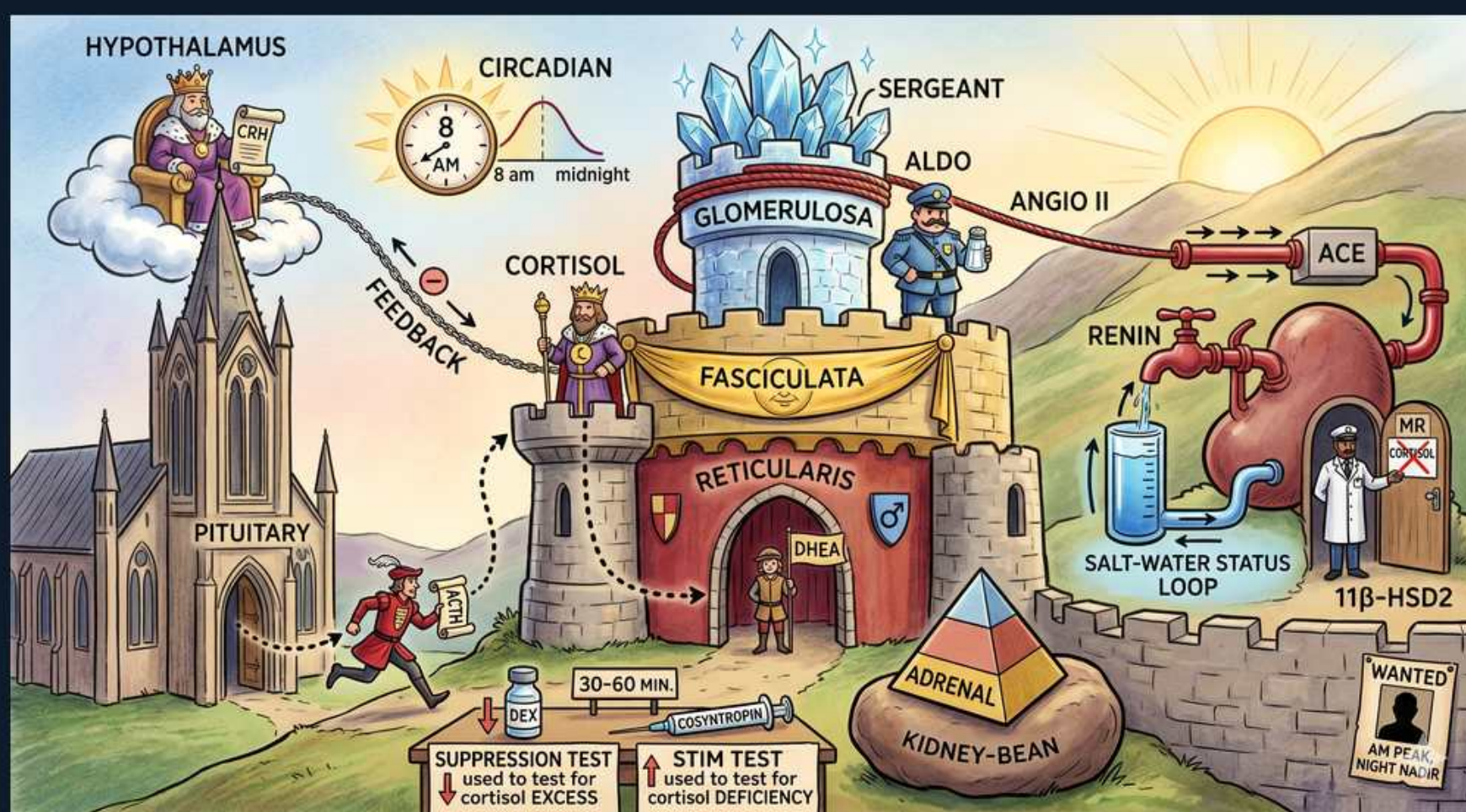


Adrenal Insufficiency/CAH — The HPA Axis & Adrenal Zones Castle



1. Hypothalamus / CRH

WHAT YOU SEE

King in the clouds over the church labeled CRH

WHAT IT ENCODES

The hypothalamus sits at the top of the HPA axis and secretes CRH (corticotropin-releasing hormone) in a circadian, pulsatile fashion, peaking in the early morning. CRH travels down the hypophyseal portal system to the anterior pituitary. Loss of CRH (or its chronic suppression by exogenous steroids) produces tertiary adrenal insufficiency. Stress, hypoglycemia, and cytokines override the normal rhythm to drive CRH up.

BOARD TRAP

Tertiary (hypothalamic) AI from chronic exogenous glucocorticoids is the single most common cause of adrenal insufficiency overall.

2. Pituitary / ACTH

WHAT YOU SEE

Jester running from the church labeled pituitary

WHAT IT ENCODES

The anterior pituitary corticotrophs respond to CRH by cleaving the precursor POMC into ACTH (and MSH). ACTH then drives cortisol and adrenal androgen synthesis, mainly by upregulating the rate-limiting StAR-mediated cholesterol delivery and steroidogenic enzymes. Pituitary destruction (tumor, Sheehan, apoplexy) lowers ACTH → secondary AI. Because POMC also yields MSH, very high ACTH in primary AI causes skin hyperpigmentation.

BOARD TRAP

ACTH does NOT control aldosterone — only cortisol and androgens — so pituitary disease spares the mineralocorticoid axis.

3. Cortisol feedback

WHAT YOU SEE

Crowned cortisol figure with dashed negative-feedback arrow back up

WHAT IT ENCODES

Cortisol exerts negative feedback on both the hypothalamus (CRH) and pituitary (ACTH), closing the loop. When cortisol falls (primary AI, CAH 21-hydroxylase block), feedback is lost and ACTH/CRH rise dramatically. When cortisol is given exogenously or autonomously secreted (Cushing), feedback suppresses ACTH. This feedback principle underlies all dynamic testing: you stimulate a deficient axis and suppress an excess axis.

BOARD TRAP

In CAH the enzyme block lowers cortisol → unopposed ACTH drive shunts precursors into androgens; treat by giving glucocorticoid to restore feedback and shut off the ACTH drive.

4. 8 AM circadian peak

WHAT YOU SEE

Clock at 8 AM with circadian cortisol curve

WHAT IT ENCODES

Cortisol follows a diurnal rhythm, peaking around 8 AM and reaching a nadir near midnight. This is why screening for deficiency uses an 8 AM cortisol: <3 µg/dL is essentially diagnostic of AI, while >18 µg/dL effectively excludes it; intermediate values need a cosyntropin stimulation test. Conversely, a LOSS of the midnight nadir (elevated late-night cortisol) is used to screen for Cushing.

BOARD TRAP

A LOW morning (8 AM) cortisol screens for deficiency; a HIGH late-night/midnight cortisol screens for excess (Cushing) — don't mix the timing.

5. Zona glomerulosa

WHAT YOU SEE

Top castle layer GLOMERULOSA with Aldo sergeant

WHAT IT ENCODES

Outermost cortical layer (the 'Salt' of GFR → Salt, Sugar, Sex) makes aldosterone. It is regulated by the renin-angiotensin system and potassium, NOT by ACTH. Aldosterone acts on the distal nephron to reabsorb Na⁺ (retaining water/volume) and excrete K⁺ and H⁺. Its loss in primary AI causes hyponatremia, hyperkalemia, salt craving, and volume depletion/orthostasis.

BOARD TRAP

Because aldosterone is RAAS-driven, only PRIMARY AI loses it; secondary/tertiary AI keeps aldosterone and therefore normal potassium.

6. Zona fasciculata

WHAT YOU SEE

Middle yellow castle layer FASCICULATA

WHAT IT ENCODES

Middle and thickest cortical layer (the 'Sugar') makes cortisol (glucocorticoid) under ACTH control. Cortisol raises glucose (gluconeogenesis), maintains vascular tone/blood pressure (permissive for catecholamines), suppresses inflammation, and supports stress responses. Its deficiency causes fatigue, hypoglycemia, hypotension, anorexia, and (via loss of cortisol's tonic ADH suppression) dilutional hyponatremia.

BOARD TRAP

Cortisol deficiency raises ADH, so hyponatremia occurs in BOTH primary and secondary AI — hyponatremia alone cannot localize the lesion.

7. Zona reticularis / DHEA

WHAT YOU SEE

Inner castle layer RETICULARIS with DHEA figure

WHAT IT ENCODES

Innermost cortical layer (the 'Sex') makes adrenal androgens, chiefly DHEA and DHEA-S, also under ACTH control. A low DHEA-S supports the diagnosis of AI, and androgen loss explains reduced libido and loss of axillary/pubis hair in women (men get androgens mainly from the testes). DHEA-S is essentially adrenal-specific, useful when distinguishing adrenal from ovarian androgen excess too.

BOARD TRAP

Adrenal androgen loss is symptomatic mostly in women; in CAH the SAME ACTH drive instead overproduces androgens, causing virilization.

8. Renin–angiotensin tap

WHAT YOU SEE

ACE pipe and renin faucet feeding angiotensin II to Aldo

WHAT IT ENCODES

Renin (released by the juxtaglomerular apparatus in response to low volume/low Na/sympathetic tone) converts angiotensinogen to angiotensin I, then ACE makes angiotensin II, which stimulates the glomerulosa to release aldosterone. In primary AI the adrenal can't respond, so renin is HIGH and aldosterone LOW — a key confirmatory pair. This whole loop is independent of the pituitary.

BOARD TRAP

Secondary/tertiary AI spares aldosterone because RAAS — not ACTH — drives it, so renin/aldosterone are normal centrally.

9. Salt-water status loop

WHAT YOU SEE

Blue water tank labeled salt-water status loop

WHAT IT ENCODES

Aldosterone secretion is tuned to volume/sodium status and serum potassium: hypovolemia and hyperkalemia stimulate it, while volume expansion suppresses it. This is why measuring renin and aldosterone together (and checking K+) localizes the mineralocorticoid defect in primary AI. Mineralocorticoid replacement (fludrocortisone) is titrated using blood pressure, orthostasis, sodium, potassium, and renin.

BOARD TRAP

Only primary AI needs fludrocortisone; adding mineralocorticoid in secondary AI is unnecessary because aldosterone is already intact.

10. 11-beta-HSD2

WHAT YOU SEE

Chemist labeled 11-beta-HSD2 guarding the mineralocorticoid receptor

WHAT IT ENCODES

The mineralocorticoid receptor binds cortisol and aldosterone with equal affinity, but in aldosterone-target tissues the enzyme 11-beta-HSD2 inactivates cortisol to cortisone, leaving aldosterone to act. When 11-beta-HSD2 is overwhelmed or inhibited, cortisol floods the MR causing apparent mineralocorticoid excess (hypertension, hypokalemia, metabolic alkalosis). Classic inhibitors: licorice (glycyrrhetic acid) and the genetic syndrome of apparent mineralocorticoid excess (AME).

BOARD TRAP

Licorice-induced or hereditary 11-beta-HSD2 loss mimics hyperaldosteronism but with LOW renin AND LOW aldosterone (cortisol is the culprit).

11. Suppression test (DEX)

WHAT YOU SEE

Left sign SUPPRESSION TEST with DEX, for cortisol excess

WHAT IT ENCODES

The dexamethasone suppression test probes for cortisol EXCESS (Cushing): give dexamethasone and a normal axis suppresses morning cortisol; failure to suppress suggests autonomous cortisol production. It is the wrong test for deficiency. Dexamethasone is chosen because it is potent and does not cross-react in the cortisol assay.

BOARD TRAP

Never use a suppression test to diagnose adrenal insufficiency — suppress for excess, stimulate for deficiency.

12. Stim test (cosyntropin)

WHAT YOU SEE

Right sign STIM TEST with cosyntropin, 30-60 min

WHAT IT ENCODES

The cosyntropin (synthetic ACTH) stimulation test diagnoses cortisol DEFICIENCY: give 250 µg cosyntropin and measure cortisol at 30–60 minutes; a peak <18 µg/dL indicates AI. In primary AI the destroyed adrenal cannot respond. In secondary AI, if the gland has had time to atrophy it also won't respond, but in ACUTE/early secondary AI the adrenal is still primed and may respond normally (false negative).

BOARD TRAP

Recent-onset (e.g., post-pituitary-surgery, acute Sheehan) secondary AI can give a falsely NORMAL cosyntropin response before the adrenal atrophies.

13. Adrenal = kidney-bean

WHAT YOU SEE

Bean-shaped adrenal pyramid on the kidney

WHAT IT ENCODES

The adrenal gland caps the upper pole of each kidney; the cortex (three steroid-producing zones) surrounds the medulla (catecholamines). Remember GFR → Salt, Sugar, Sex: Glomerulosa-aldosterone, Fasciculata-cortisol, Reticularis-androgens, with ACTH dependence increasing as you move inward. Imaging the adrenals (CT) is part of the workup for antibody-negative primary AI (TB, hemorrhage, metastasis, infiltration).

BOARD TRAP

'The deeper you go, the more ACTH-dependent' — glomerulosa (outer) is ACTH-independent; fasciculata and reticularis (inner) are ACTH-driven.